

Skills Summary

Handling the Ambiguous SSA Case Completely

Use this reference while completing your worksheet or when studying.

Skill 1 — The height test

Given side a , side b and the angle A opposite a (SSA), the perpendicular from C to the base line has length $h = b \sin A$. Compare a with it:

- $a < h$: no triangle — the swinging side never reaches the line;
- $a = h$: one right triangle; $h < a < b$: **two** triangles;
- $a \geq b$: one triangle.

Example: $b = 14$, $A = 38^\circ$. Find the height h that decides the case.

Step 1. Before solving anything I need the perpendicular distance from the far vertex to the base line, because that is the length the swinging side has to beat.

Step 2. $h = b \sin A = 14 \sin 38^\circ = 14(0.6157) \Rightarrow h \approx \mathbf{8.62}$, so any a below 8.62 gives no triangle at all.

Try it: $b = 9$, $A = 55^\circ$. Find h .

check: $h \approx 7.37$

Skill 2 — When only one triangle fits

If $a \geq b$, the side opposite the given angle is the longer one, so the angle B opposite b must be *smaller* than A — and therefore acute. The obtuse partner is impossible and is discarded without further thought.

$$\sin B = \frac{b \sin A}{a}, \text{ then } C = 180^\circ - A - B.$$

Example: $a = 16$, $b = 10$, $A = 48^\circ$. Find B .

Step 1. Here a is bigger than b , so one triangle exists and the angle opposite the shorter side has to be acute — no second case to carry.

Step 2. $\sin B = \frac{10 \sin 48^\circ}{16} = 0.4645 \Rightarrow B \approx \mathbf{27.68^\circ}$

Try it: $a = 19$, $b = 12$, $A = 41^\circ$. Find B .

check: $B \approx 24.48^\circ$

Skill 3 — When two triangles fit

If $h < a < b$, the swinging side crosses the base line twice and there are *two* answers. The calculator returns only the acute one.

$$B_1 = \sin^{-1}\left(\frac{b \sin A}{a}\right), \quad B_2 = 180^\circ - B_1.$$

Keep B_2 only if $A + B_2 < 180^\circ$ — in this case it always is.

Example: $a = 7$, $b = 9$, $A = 47^\circ$. Find both values of B .

Step 1. The height is $9 \sin 47^\circ = 6.58$, and $6.58 < 7 < 9$, so this is the two-triangle case and stopping at one answer would report only part of it.

Step 2. $\sin B = \frac{9 \sin 47^\circ}{7} = 0.9403$, so $B_1 = 70.10^\circ$ and $B_2 = 180^\circ - 70.10^\circ \Rightarrow B = \mathbf{70.10^\circ}$ or $\mathbf{109.90^\circ}$.

Try it: $a = 10$, $b = 13$, $A = 40^\circ$. Find both values of B .

check: $B \approx 89.95^\circ$ or 123.32°

Skill 4 — Reporting both cases in context

In a real measurement, two triangles mean the data genuinely does not decide the answer. Report both, and say what extra information would settle it (a further distance, or whether the unknown angle is acute).

Discarding the obtuse case because it "looks wrong" is the commonest way to lose half the answer.

Example: A station at A sees a boat at C , $b = 300$ m away; the boat is $a = 250$ m from a marker at B on the shoreline, and $A = 49^\circ$. Find both possible angles at the marker.

Step 1. The height is $300 \sin 49^\circ = 226.4$ m and $226.4 < 250 < 300$, so the measurements fit two boat positions and both must be reported.

Step 2. $\sin B = \frac{300 \sin 49^\circ}{250} = 0.9055$, so $B_1 = 64.91^\circ$ and $B_2 = 180^\circ - 64.91^\circ \Rightarrow B = \mathbf{64.91^\circ}$ or $\mathbf{115.09^\circ}$.

Try it: Same setup with $b = 220$ m, $a = 180$ m and $A = 46^\circ$. Find both possible angles at the marker.

check: $B \approx 61.55^\circ$ or 118.45°